GEOMETRY – SHEET 7 – Surface Area and Isometry

1. Let $\mathbf{r}$ be a parametrization of a subset of the xy-plane in $\mathbb{R}^3$ so that

$$\mathbf{r}(u,v) = (x(u,v), y(u,v), 0), \qquad (u,v) \in U.$$

Determine $\mathbf{r}_u \wedge \mathbf{r}_v$ and show that the area of $\mathbf{r}(U)$ equals

$$\iint_U \left| \frac{\partial(x,y)}{\partial(u,v)} \right| \mathrm{d}u\,\mathrm{d}v.$$

2. Part of a catenoid is formed by rotating the graph of $y = \cosh x$ (where $a \leqslant x \leqslant b$) about the x-axis. Calculate the area of the resulting surface.

3. Let $0 < \alpha < \pi/2$ and let S_α be the cap of the unit sphere $x^2 + y^2 + z^2 = 1$ parametrized as

$$\mathbf{r}(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta) \qquad 0 < \theta < \alpha,\ 0 < \phi < 2\pi.$$

(i) Determine $\mathbf{r}_\theta \wedge \mathbf{r}_\phi$ and show that the surface area of S_α equals $2\pi(1 - \cos\alpha)$.

(ii) Rederive the result of (i) by considering the cap as part of the graph $z = \sqrt{1 - x^2 - y^2}$ and using the formula for a graph's surface area.

4. Let θ and ϕ denote spherical polar co-ordinates on the unit sphere $x^2 + y^2 + z^2 = 1$. The *Albers equal-area conic projection* is defined by

$$x = \frac{1}{n}\sqrt{C - 2n\cos\theta}\,\sin n\phi, \qquad y = \rho_0 - \frac{1}{n}\sqrt{C - 2n\cos\theta}\,\cos n\phi,$$

where n, C, ρ_0 are constants. Describe the image in the plane of a latitude ($\theta = $ const.) and a meridian ($\phi = $ const.)

Determine the Jacobian $|\partial(x,y)/\partial(\phi,\theta)|$ and explain why this means the projection preserves area.

5. (i) Let $\mathbf{r}(u,v)$ be a parametrization of a surface in $\mathbb{R}^3$ and $\boldsymbol{\gamma}(t) = \mathbf{r}(u(t), v(t))$. Show the following:

$$\boldsymbol{\gamma}'(t) = u'\mathbf{r}_u + v'\mathbf{r}_v, \qquad |\boldsymbol{\gamma}'(t)|^2 = E(u')^2 + 2Fu'v' + G(v')^2, \qquad |\mathbf{r}_u \wedge \mathbf{r}_v|^2 = EG - F^2,$$

where $E = \mathbf{r}_u \cdot \mathbf{r}_u$, $F = \mathbf{r}_u \cdot \mathbf{r}_v$, $G = \mathbf{r}_v \cdot \mathbf{r}_v$.

(ii) The *flat torus* in $\mathbb{R}^4$ can be parametrized as

$$\mathbf{r}(\theta,\phi) = (\cos\theta, \sin\theta, \cos\phi, \sin\phi), \qquad (\theta,\phi) \in U = (0,2\pi) \times (0,2\pi).$$

Let $\gamma(t) = (\theta(t), \phi(t))$ be a curve in U. Show that the curve γ, and its image $\mathbf{r}(\gamma)$ in the flat torus, have the same length.

As there is no vector product in $\mathbb{R}^4$ we cannot use our current formulae to work out the surface area of the flat torus. What do you think its surface area equals?

6. Compute the area of the helicoid

$$\mathbf{s}(X,Z) = (X\cos Z, X\sin Z, Z)$$

for $A < X < B$, $0 < Z < C < 2\pi$.

Rederive the result by defining an isometry between the catenoid and helicoid and using the calculation in Question 2.

You may find it helpful to use the identities: $\sinh(2x) = 2\sinh x\cosh x$, $\cosh(\operatorname{arcsinh} x) = \sqrt{1 + x^2}$.

7. (*Optional*) Let $0 < a \leqslant b \leqslant c < \pi/2$ and let $A = (0,0,1)$ and $B = (\sin c, 0, \cos c)$ be points on the unit sphere.

(i) Show that there is a point C which is at a distance b from A and at a distance a from B if and only if $a + b \geqslant c$.

[Hint: every point at a distance b from A has the form $(\sin b \cos \phi, \sin b \sin \phi, \cos b)$ for some ϕ.]

(ii) Prove the *spherical cosine rule* which states that for a spherical triangle with sides a, b, c opposite angles α, β, γ

$$\cos a = \cos b \cos c + \sin b \sin c \cos \alpha.$$

Show that when a, b, c are small enough, that we can ignore terms of order 3 and above, so that the approximations $\cos x \approx 1 - x^2/2$ and $\sin x \approx x$ apply, then the spherical cosine rule approximates the usual cosine rule.